

Algoritmo

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R Markdown

This is an R Markdown document. Markdown is a simple formatting syntax for authoring HTML, PDF, and MS Word documents. For more details on using R Markdown see <http://rmarkdown.rstudio.com>.

When you click the **Knit** button a document will be generated that includes both content as well as the output of any embedded R code chunks within the document. You can embed an R code chunk like this:

Primeros comandos de R

R es un lenguaje muy similar a Matlab y permite el trabajo con matrices

```
A=38
A=40
B=60
C=B-A
C
```

```
## [1] 20
```

```
Y=40
alpha<-30
```

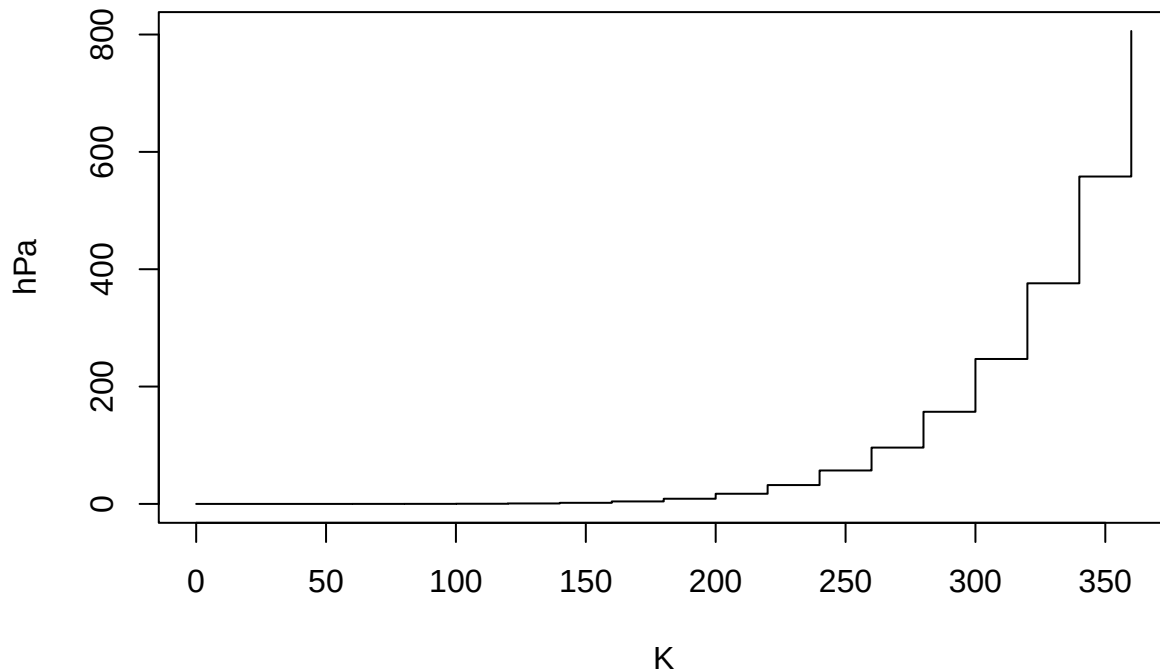
Uso de datasets o base de datos internos de R

Usando el comando `data` en la consola obtenemos distintos datos ya cargados en la base de datos R, estos datos estan dados en tablas de las cuales podemos elegir columnas específicas si escribimos `$` después del comando de información específico y elegimos la columna de datos deseada `## Including Plots`

You can also embed plots, for example:

```
plot(pressure,main="presión del gas ideal",type="s",ylab="hPa",xlab="K")
```

presión del gas ideal



Comando Plot

Es un comando que permite graficar cualquier fuente de datos que le asignemos, teniendo también la posibilidad de asignar nombre a las variables o hace cambios al gráfico. En caso de duda de como utilizar el comando podemos escribir `?plot` en la consola, lo cual abrirá una ventana que explicará en detalle el comando.

Funciones de estadísticas

```
z1<-rnorm(350,22,5)
z1
```

```
## [1] 27.175996 28.503795 22.726816 24.597777 22.045372 22.126832 37.469336
## [8] 18.824387 24.850544 16.598199 22.300488 19.616765 23.725338 28.515186
## [15] 20.605807 28.885465 14.942358 26.287571 20.063999 16.620169 16.541680
## [22] 26.480139 23.499659 19.019836 17.345933 21.101276 12.047922 23.499173
## [29] 17.049299 26.449513 20.216705 23.079257 32.777293 23.513670 18.863575
## [36] 30.379927 20.102039 12.996015 21.907816 22.211902 19.558058 22.468742
## [43] 21.405009 18.146962 23.189205 25.147234 25.628495 19.270461 25.518289
## [50] 20.274457 28.039058 12.008120 20.804945 19.026023 19.498995 23.730425
## [57] 21.637745 16.517432 12.012786 14.160933 20.640952 20.586758 25.011495
## [64] 18.581644 23.444227 22.664642 20.646027 14.132441 29.208648 19.534993
## [71] 34.822217 20.456935 13.963103 16.549446 23.012996 12.068659 22.017982
## [78] 17.621867 20.601098 25.924296 14.676999 28.321628 23.476545 13.717652
## [85] 28.289839 26.515087 30.007048 34.248524 21.715988 31.254535 22.008893
## [92] 26.102942 25.366603 28.771964 25.702210 24.210595 21.255426 17.259103
## [99] 23.207594 23.317334 21.832722 20.829763 31.387973 24.045100 18.950499
## [106] 18.970290 16.004657 21.838883 28.531043 22.710585 25.564611 20.307418
## [113] 24.128612 28.761554 35.195734 38.286327 29.243080 18.008069 20.668988
```

```
## [120] 36.416972 27.238892 13.537014 20.567252 23.956766 26.683713 16.660405
## [127] 28.475582 19.385316 18.203258 18.329594 29.113400 26.212686 20.719733
## [134] 21.767397 11.772876 22.950567 24.543115 27.596292 28.774770 29.439017
## [141] 28.951774 21.832727 19.115852 18.257727 19.110038 26.128599 21.048298
## [148] 19.467398 10.081356 16.950418 22.420753 23.601841 22.275384 24.403341
## [155] 15.887616 21.194694 14.913696 14.124948 16.277001 16.880866 30.822606
## [162] 23.424131 21.350448 9.289400 22.677173 26.341650 21.691946 8.336674
## [169] 18.763201 16.258473 18.225257 17.036910 28.173497 12.875723 26.913244
## [176] 18.770464 29.650051 23.526513 25.362619 28.848164 18.729322 22.328881
## [183] 24.385840 22.419743 20.634563 29.174419 25.518562 22.779744 18.527016
## [190] 25.069447 25.033023 15.155056 18.953377 24.100361 23.285528 34.399573
## [197] 26.312628 35.345714 6.873835 18.271954 14.042014 30.348971 19.946315
## [204] 20.924298 30.481413 27.033942 20.904278 13.888923 24.401888 18.701706
## [211] 19.998297 30.975427 22.166839 23.947374 12.315765 13.395625 26.340109
## [218] 16.566366 11.815037 18.494009 22.625152 28.049579 11.807845 21.303340
## [225] 18.854489 19.797936 22.556396 33.052190 18.553035 23.243565 16.424562
## [232] 21.942662 19.672070 24.345542 32.837085 19.848214 28.274812 20.264453
## [239] 20.337087 25.387680 29.772260 23.247651 27.571551 24.572204 23.083238
## [246] 13.641767 20.317859 18.777118 20.223645 26.874124 31.067013 23.448534
## [253] 15.200438 28.508191 17.389352 21.629200 9.627902 22.696432 14.988276
## [260] 21.247207 18.491970 28.255102 15.137691 7.294061 23.212388 25.065210
## [267] 26.198759 25.228183 10.937906 29.552555 26.979032 23.434765 20.431785
## [274] 17.269903 12.407378 21.456424 22.083231 19.258125 16.163037 23.504150
## [281] 17.863348 20.956671 21.464239 22.485888 23.094057 15.203135 16.406252
## [288] 29.038451 24.851241 22.021444 26.755418 32.580294 22.802037 21.741456
## [295] 27.973945 27.345351 22.145031 23.955119 8.794711 24.863167 20.911495
## [302] 15.669756 21.928186 26.975615 21.993217 26.780114 16.338307 18.232095
## [309] 21.950753 18.228171 20.036491 19.113418 18.499909 25.284051 21.110465
## [316] 9.045901 23.061765 23.952392 31.367008 26.171977 19.301163 13.919212
## [323] 20.555224 25.637332 22.116168 23.932959 21.541762 26.003892 27.668738
## [330] 18.707568 22.179404 14.706426 25.964440 15.338074 25.531941 14.788709
## [337] 34.244633 22.662073 28.316158 20.383337 22.273399 20.569575 22.801882
## [344] 16.581463 14.466743 10.811794 19.933260 28.379458 20.526037 28.586313
```

350 es la cantidad de valores 22 es el promedio de valores 5 es la desviación estandar

```
#comentario
w1<-length(z1)
w1
```

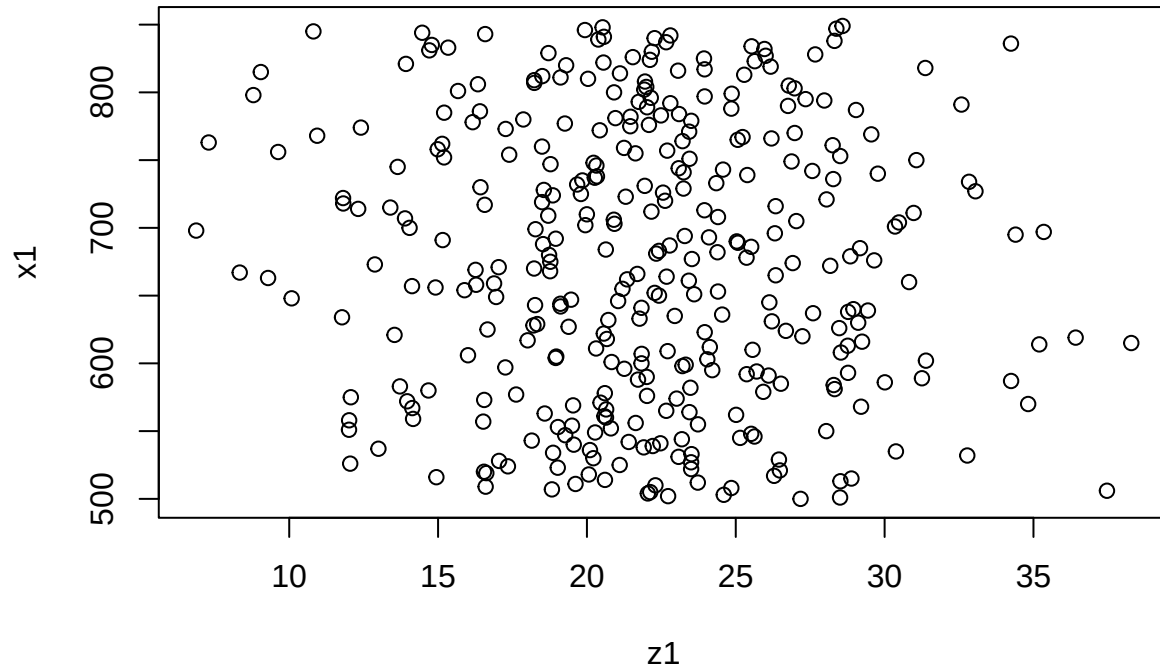
```
## [1] 350
```

```
x1<-(500:849)
x1
```

```
## [1] 500 501 502 503 504 505 506 507 508 509 510 511 512 513 514 515 516 517
## [19] 518 519 520 521 522 523 524 525 526 527 528 529 530 531 532 533 534 535
## [37] 536 537 538 539 540 541 542 543 544 545 546 547 548 549 550 551 552 553
## [55] 554 555 556 557 558 559 560 561 562 563 564 565 566 567 568 569 570 571
## [73] 572 573 574 575 576 577 578 579 580 581 582 583 584 585 586 587 588 589
## [91] 590 591 592 593 594 595 596 597 598 599 600 601 602 603 604 605 606 607
## [109] 608 609 610 611 612 613 614 615 616 617 618 619 620 621 622 623 624 625
## [127] 626 627 628 629 630 631 632 633 634 635 636 637 638 639 640 641 642 643
## [145] 644 645 646 647 648 649 650 651 652 653 654 655 656 657 658 659 660 661
## [163] 662 663 664 665 666 667 668 669 670 671 672 673 674 675 676 677 678 679
## [181] 680 681 682 683 684 685 686 687 688 689 690 691 692 693 694 695 696 697
```

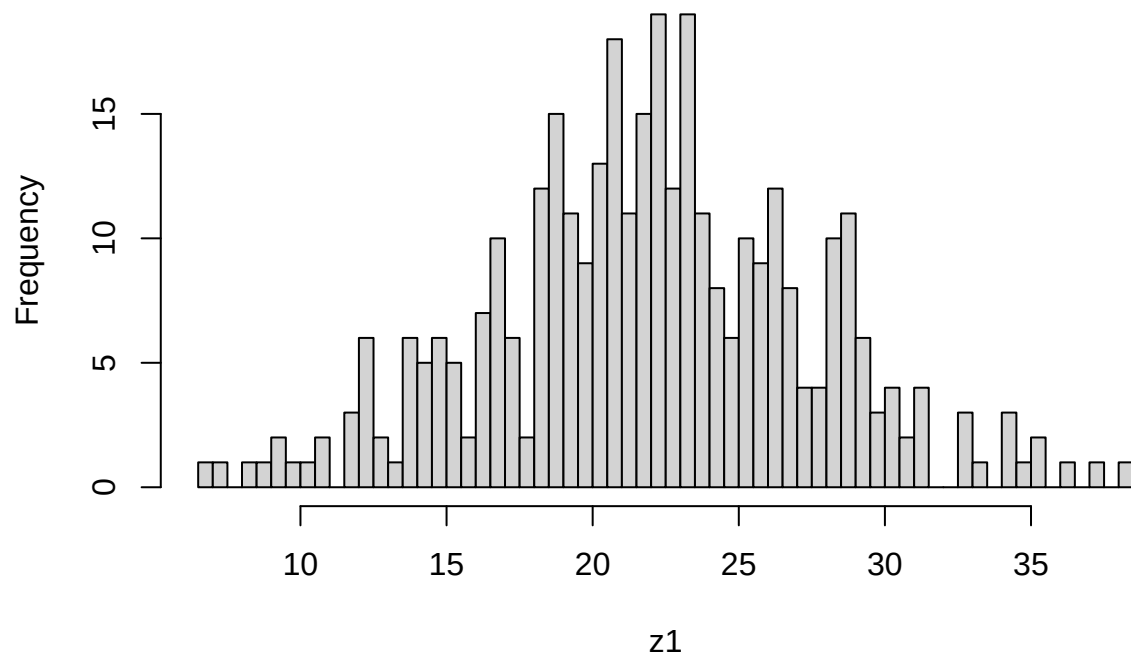
```
## [199] 698 699 700 701 702 703 704 705 706 707 708 709 710 711 712 713 714 715
## [217] 716 717 718 719 720 721 722 723 724 725 726 727 728 729 730 731 732 733
## [235] 734 735 736 737 738 739 740 741 742 743 744 745 746 747 748 749 750 751
## [253] 752 753 754 755 756 757 758 759 760 761 762 763 764 765 766 767 768 769
## [271] 770 771 772 773 774 775 776 777 778 779 780 781 782 783 784 785 786 787
## [289] 788 789 790 791 792 793 794 795 796 797 798 799 800 801 802 803 804 805
## [307] 806 807 808 809 810 811 812 813 814 815 816 817 818 819 820 821 822 823
## [325] 824 825 826 827 828 829 830 831 832 833 834 835 836 837 838 839 840 841
## [343] 842 843 844 845 846 847 848 849
```

```
plot(z1,x1)
```



```
hist(z1,main="histograma de edades", breaks=50)
```

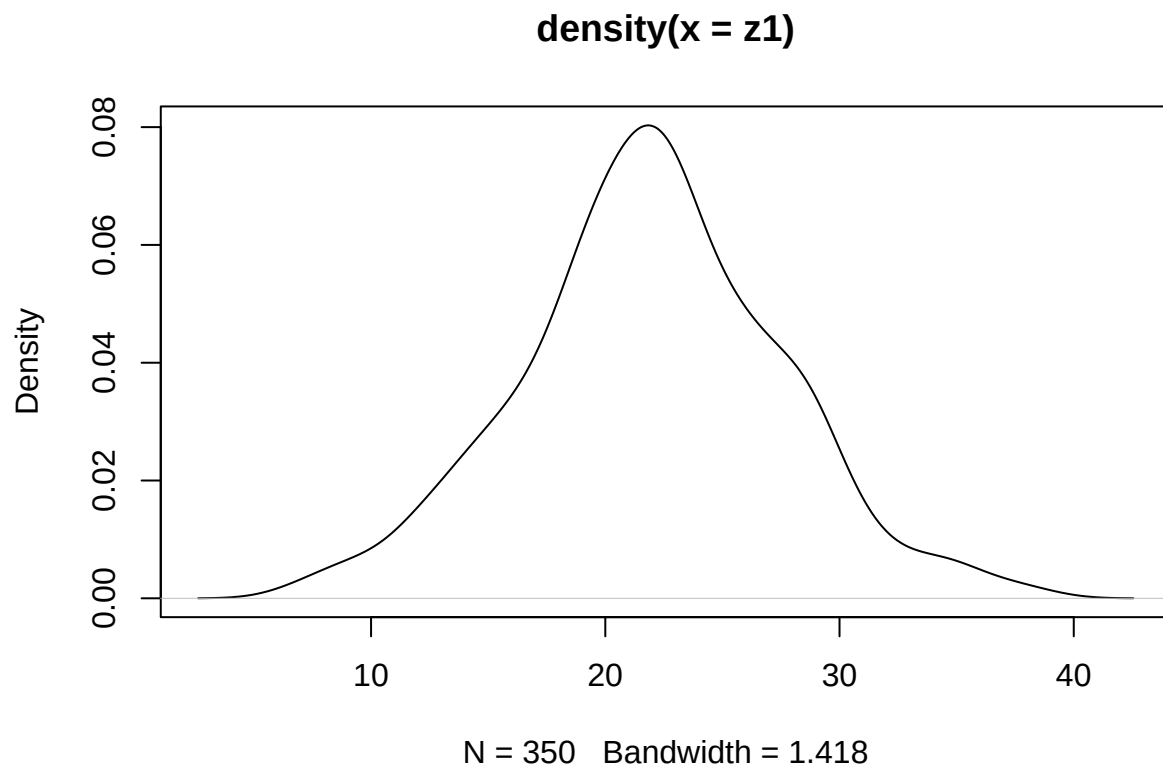
histograma de edades



```
density(z1)
```

```
##  
## Call:  
## density.default(x = z1)  
##  
## Data: z1 (350 obs.); Bandwidth 'bw' = 1.418  
##  
##      x          y  
## Min.   : 2.618   Min.   :1.041e-05  
## 1st Qu.:12.599   1st Qu.:3.386e-03  
## Median :22.580   Median :1.359e-02  
## Mean   :22.580   Mean    :2.500e-02  
## 3rd Qu.:32.561   3rd Qu.:4.334e-02  
## Max.   :42.542   Max.    :8.030e-02
```

```
plot(density(z1),type="l")
```



Note that the `echo = FALSE` parameter was added to the code chunk to prevent printing of the R code that generated the plot.